

# 1 Comparison Test of Nonlinear Equation-Solving Capabilities: 1stOpt, Matlab, Mathematica, and Maple

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## 1.1 Introduction

Solving systems of equations is one of the most fundamental and essential components of scientific numerical computation. Many scientific computing problems eventually transform into the problem of solving systems of equations. For linear systems of equations, the process is relatively straightforward, as efficient and reliable algorithms can guarantee a correct solution. However, for nonlinear systems of equations, significant challenges remain, as no algorithm or software tool can ensure a guaranteed correct solution.

Matlab, Mathematica, and Maple are three of the most well-known scientific numerical computation software, also known as the '3M' (previously referred to as '4M' when MathCAD was included). Among the '3M' software, Matlab excels in numerical computation and is the most widely used, whereas Mathematica and Maple are renowned for their strengths in symbolic reasoning and computation, while still providing support for numerical tasks. All three software packages come with built-in functionalities and commands for solving nonlinear systems of equations, such as 'fsolve' and 'vpasolve' in Matlab, 'NSolve' and 'FindRoot' in Mathematica, and 'fsolve' in Maple.

Solving nonlinear systems of equations is also one of the key functionalities of 1stOpt. This raises several questions regarding the nonlinear equation-solving capabilities of the '3M' software: their effectiveness, ease of use, and how they compare with 1stOpt in terms of strengths and weaknesses. To provide an initial perspective, a preliminary comparative evaluation is conducted based on twelve practical cases of nonlinear systems of equations.



## 1.2 Case Studies and Computational Comparison Test

Twelve nonlinear equation systems arising from practical applications were selected. These systems are relative standard in form, involving mainly commonly used exponential and logarithmic functions, without any special functions or even trigonometric terms. The dimensionality of the systems is also modest, ranging from three to eight variables.

In this comparison, the focus is not only on the correctness of the solutions but also on the conciseness of the code required to solve each problem. This helps illustrate the practical usability of different software tools when handling nonlinear equation systems.

Among the '3M' software, only Matlab's 'fsolve,' Mathematica's 'FindRoot,' and Maple's 'fsolve' were tested, as these are the most representative built-in methods for solving systems of nonlinear equations in each environment. Since the computation results of these commands depend on the initial values, random

initial points within the range of  $[-10, 10]$  were generated for all tests to ensure the consistency. In contrast, 1stOpt does not require users to manually assigned initial values, as the software automatically generates them internally.

All four software were evaluated under comparable conditions: MATLAB R2024b, Mathematica 14.3(free trial), Maple 2026(free trial), and 1stOpt 11.0. The tests were conducted on a computer with an Intel Core i7-12700K CPU and 32GB of RAM. Each solver was run independently ten times. The correctness of the obtained solutions was assessed based on equation-error precision, typically requiring the residual (the difference between the left and right sides of the equations) to be on the order of  $1\text{E-}10$  or smaller.

If you are aware of alternative commands or settings in these software environments that may yield better results, please feel free to share them in the comments or join our Discord channel (<https://discord.gg/pyNyuQNBd>) for further discussion! And this isn't limited to the 3M software - if you have achieved similar or better performance using other tools or methods, we'd love to hear about it!

### 1.2.1 Case I: A three-equation system with three variables

$$\begin{cases} (1 - x_1) \cdot 0.7133 - (1 - 0.7133) = x_1 \cdot \left(1 - \exp\left(-\frac{(1 - x_1) \cdot 127}{x_2}\right)\right) \cdot \exp\left(-\frac{127}{x_3}\right) \\ (1 - x_1) \cdot 0.8058 + (1 - 0.8058) = x_1 \cdot \left(1 - \exp\left(-\frac{(1 - x_1) \cdot 229}{x_2}\right)\right) \cdot \exp\left(-\frac{229}{x_3}\right) \\ (1 - x_1) \cdot 0.7133 - (1 - 0.8708) = x_1 \cdot \left(1 - \exp\left(-\frac{(1 - x_1) \cdot 421}{x_2}\right)\right) \cdot \exp\left(-\frac{421}{x_3}\right) \end{cases}$$

Case I - Code Solution

1stOpt Code Solution:

```
Hardness = 2;
(1-x1)*0.7133-(1-0.7133) = x1*(1-exp(-(1-x1)*127/x2))*exp(-127/x3);
(1-x1)*0.8058+(1-0.8058) = x1*(1-exp(-(1-x1)*229/x2))*exp(-229/x3);
(1-x1)*0.7133-(1-0.8708) = x1*(1-exp(-(1-x1)*421/x2))*exp(-421/x3);
```

Matlab Code Solution:

```
equations = @(x) [(1-x(1))*0.7133-(1-0.7133)-(x(1)*(1-exp(0-(1-x(1))*127/x(2)))*exp(0-127/x(3)))
);
(1-x(1))*0.8058+(1-0.8058)-(x(1)*(1-exp(0-(1-x(1))*229/x(2)))*exp(0-229/x(3)))
(1-x(1))*0.7133-(1-0.8708)-(x(1)*(1-exp(0-(1-x(1))*421/x(2)))*exp(0-421/x(3)))];
InitialGuess = 10*(2*rand(3,1)-1); % [1,1,1];
%InitialGuess = ones(3,1); % [1,1,1];
options = optimoptions('fsolve','Display','iter','MaxFunEvals',5000);
[x,fval] = fsolve(equations,InitialGuess,options);
```

Mathematica Code Solution:

```
FindRoot[{(1-x1)*0.7133-(1-0.7133)-(x1*(1-Exp[0-(1-x1)*127/x2])*Exp[0-127/x3]) == 0,
(1-x1)*0.8058+(1-0.8058)-(x1*(1-Exp[0-(1-x1)*229/x2])*Exp[0-229/x3]) == 0,
(1-x1)*0.7133-(1-0.8708)-(x1*(1-Exp[0-(1-x1)*421/x2])*Exp[0-421/x3]) == 0},
{{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)}},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```
interface(displayprecision = 16)
with(RandomTools):
fsolve({(1-x1)*0.7133-(1-0.7133)-(x1*(1-exp(0-(1-x1)*127/x2))*exp(0-127/x3)) = 0,
(1-x1)*0.8058+(1-0.8058)-(x1*(1-exp(0-(1-x1)*229/x2))*exp(0-229/x3)) = 0,
(1-x1)*0.7133-(1-0.8708)-(x1*(1-exp(0-(1-x1)*421/x2))*exp(0-421/x3)) = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1)})
```

The results for Case I are shown in Table 1:

The three "3M" software tools were unable to obtain the correct solution for case I. In the 1stOpt code, an additional line **Hardness = 2;** is added, indicating that the problem is relatively difficult. This change the default difficulty level of from 1 to 2 (the scale ranges from 1 to 5). When the difficulty level is increased, 1stOpt automatically adjusts several internal control parameters, which increases the likelihood of finding the optimal solution, though it typically requires more computation time. With the default difficulty level of 1, the success rate is around 30%. After increasing it to level 2, the success rate rises to 90%.

Table 1: Numerical solutions and residual errors for Case I

|             | Parameter Values                                                                                                                                                | Residual values                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1stOpt      | x1: 24.6585100975876<br>x2: 2002.68074188802<br>x3: 78.8665346900753                                                                                            | -7.105427357601E-15<br>3.5527136788005E-15<br>-3.5527136788005E-15    |
| Matlab      | Either no solutions or incorrect solutions, :<br>x1 = 0.650398789596784<br>x2 = 83.6695142051881<br>x3 = 400.245114026375                                       | -0.232331524448736<br>0.249857694149897<br>-0.0678863445966185        |
| Mathematica | Each run produces a different result, but all of them are incorrect solutions. The best result is given below::<br>x1 = -2.13222<br>x2 = 9.4542<br>x3 = 9.43613 | 1.94751557180988<br>2.71814287606154<br>2.105012526                   |
| Maple       | No solution or erroneous results, such as:<br>x1 = -3.8586935440000000<br>x2 = -8.0786647770000000<br>x3 = -7.6607866820000000                                  | -9.07531432126194E40<br>-2.41075384736122E73<br>-2.60683030432902E134 |

### 1.2.2 Case II: A three-equation system with three variables

$$\begin{cases} x_1^{x_2} + x_2^{x_1} - 5 \cdot x_1 \cdot x_2 \cdot x_3 - 85 = 0 \\ x_1^3 \cdot x_2^{x_3} \cdot x_3^{x_2} - 60 = 0 \\ x_1^{x_3} + x_3^{x_1} - x_2 - 0.55 = 0 \end{cases}$$

Case II - Code Solution

1stOpt Code Solution:

```
Function x1^x2+x2^x1-5*x1*x2*x3-85=0;
        x1^3*x2^x3*x3^x2-60=0;
        x1^x3+x3^x1-x2-0.55=0;
```

Matlab Code Solution:

```
equations = @(x) [x(1)^x(2)+x(2)^x(1)-5*x(1)*x(2)*x(3)-85-0;
x(1)^3*x(2)^x(3)*x(3)^x(2)-60-0;
x(1)^x(3)+x(3)^x(1)-x(2)-0.55-0];
InitialGuess = 10*(2*rand(3,1)-1); %[1,1,1];
%InitialGuess = ones(3,1); %[1,1,1];
options = optimoptions('fsolve','Display','iter','MaxIterations',5000,'MaxFunEvals',5000);
[x,fval] = fsolve(equations,InitialGuess,options);
```

Mathematica Code Solution:

```
FindRoot[{x1^x2+x2^x1-5*x1*x2*x3-85-0 == 0,
x1^3*x2^x3*x3^x2-60-0 == 0,
x1^x3+x3^x1-x2-0.55-0 == 0},
{{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)}},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```
interface(displayprecision = 16)
with(RandomTools):
fsolve({x1^x2+x2^x1-5*x1*x2*x3-85-0 = 0,
x1^3*x2^x3*x3^x2-60-0 = 0,
x1^x3+x3^x1-x2-0.55-0 = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1)})
```

The results for Case II are shown in Table 2:

Table 2: Numerical solutions and residual errors for Case II

|             | Parameter Values                                             | Residual values                                                        |
|-------------|--------------------------------------------------------------|------------------------------------------------------------------------|
| 1stOpt      | x1: 19390.8841948434<br>x2: 0.45<br>x3: 2.33057529481649E-25 | -3.26849658449646E-13<br>1.30029320644098E-12<br>-3.99680288865056E-15 |
| Matlab      | Either no solution, or only complex solutions.               |                                                                        |
| Mathematica | Either no solution, or only complex solutions.               |                                                                        |
| Maple       | Either no solution, or only complex solutions.               |                                                                        |

1stOpt is the only one capable of obtaining the correct solution, achieving a success rate of 20% under default settings. If **Hardness** = 2; is added, the success rate increases to 50%. Although this problem involves only three parameters, it remains highly challenging to solve. Even when the three "3M" software tools are provided with initial values close to the optimal solution, such as  $x = [20000, 1, 0]$ , they still fail to find the correct solution. In contrast, when 1stOpt is given such initial values, the success rate reaches 100%.

### 1.2.3 Case III: A three-equation system with three variables

$$\begin{cases} 2 \cdot x_1 \cdot (1 - x_2 \cdot \exp(-13 \cdot x_3)) = 2.07 \\ 3 \cdot x_1 \cdot (1 - x_2 \cdot \exp(-21 \cdot x_3)) = 3.22 \\ 4 \cdot x_1 \cdot (1 - x_2 \cdot \exp(-29 \cdot x_3)) = 5.14 \end{cases}$$

Case III - Code Solution

1stOpt Code Solution:

```
Parameter x1,x2,x3;
Function 2*x1*(1-x2*exp(-13*x3))=2.07;
          3*x1*(1-x2*exp(-21*x3))=3.22;
          4*x1*(1-x2*exp(-29*x3))=5.14;
```

Matlab Code Solution:

```
equations = @(x) [2*x(1)*(1-x(2)*exp(0-13*x(3)))-2.07;
3*x(1)*(1-x(2)*exp(0-21*x(3)))-3.22;
4*x(1)*(1-x(2)*exp(0-29*x(3)))-5.14];
InitialGuess = 10*(2*rand(3,1)-1); %[1,1,1];
%InitialGuess = ones(3,1); %[1,1,1];
options = optimoptions('fsolve','Display','iter','MaxIterations',5000,'MaxFunEvals',5000);
[x,fval] = fsolve(equations,InitialGuess,options);
```

Mathematica Code Solution:

```
FindRoot[{2*x1*(1-x2*Exp[0-13*x3])-2.07 == 0,
3*x1*(1-x2*Exp[0-21*x3])-3.22 == 0,
4*x1*(1-x2*Exp[0-29*x3])-5.14 == 0},
{{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)}},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```
interface(displayprecision = 16)
with(RandomTools):
fsolve({2*x1*(1-x2*exp(0-13*x3))-2.07 = 0,
3*x1*(1-x2*exp(0-21*x3))-3.22 = 0,
4*x1*(1-x2*exp(0-29*x3))-5.14 = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1)})
```

The results for Case III are shown in Table 3:

This is a fairly ordinary low-dimensional nonlinear system of equations, yet Matlab and Mathematica are only able to produce approximate solutions. In contrast, both 1stOpt and Maple can obtain the correct solution with a 100% success rate.

Table 3: Numerical solutions and residual errors for Case III

|             | Parameter Values                                                                                                                                                           | Residual values                                                       |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1stOpt      | x1: 1.02652243590673<br>x2: -0.000514078360857846<br>x3: -0.21358660881644                                                                                                 | 1.78586034849104E-11<br>2.21276330591991E-11<br>-3.03845837379413E-12 |
| Matlab      | Either no solution is obtained, or only an approximate real solution is returned, such as:<br>x1 = 1666.77674862914<br>x2 = 0.999556379472030<br>x3 = 1.09984162407442e-05 | -0.114783221524789<br>0.152513479556321<br>-0.0571158585310094        |
| Mathematica | Either no solution is obtained, or only an approximate real solution is returned, such as:<br>$5.386756645619156 * 10^{-59}$<br>-4.331007849135506<br>-4.584604671108670   | -2.07<br>-3.22<br>1.68753899743024E-14                                |
| Maple       | x1 = 1.0265224360000000<br>x2 = -0.0005140783609000<br>x3 = -0.2135866088000000                                                                                            | 2.03705941004273E-10<br>2.7773028321576E-10<br>5.58344481760287E-11   |

### 1.2.4 Case IV: A three-equation system with three variables

$$\begin{cases} \frac{x_1}{1+x_2 \cdot (2.72^{-x_3 \cdot 1990})} - \frac{x_1}{1+x_2} = 0.22 \\ \frac{x_1}{1+x_2 \cdot (2.72^{-x_3 \cdot 2002})} - \frac{x_1}{1+x_2} = 0.31 \\ \frac{x_1}{1+x_2 \cdot (2.72^{-x_3 \cdot 2005})} - \frac{x_1}{1+x_2} = 0.33 \end{cases}$$

Case IV - Code Solution

1stOpt Code Solution:

```
Function
x1/(1+x2*(2.72^(-x3*1990)))-x1/(1+x2)=0.22;
x1/(1+x2*(2.72^(-x3*2002)))-x1/(1+x2)=0.31;
x1/(1+x2*(2.72^(-x3*2005)))-x1/(1+x2)=0.33;
```

Matlab Code Solution:

```
equations = @(x) [x(1)/(1+x(2)*(2.72^(-x(3)*1990)))-x(1)/(1+x(2))-0.22;
x(1)/(1+x(2)*(2.72^(-x(3)*2002)))-x(1)/(1+x(2))-0.31;
x(1)/(1+x(2)*(2.72^(-x(3)*2005)))-x(1)/(1+x(2))-0.33];
InitialGuess = 10*(2*rand(3,1)-1); % [1,1,1];
%InitialGuess = ones(3,1); % [1,1,1];
options = optimoptions('fsolve','Display','iter','MaxIterations',5000,'MaxFunEvals',5000);
[x,fval] = fsolve(equations,InitialGuess,options);
```

Mathematica Code Solution:

```
FindRoot[{x1/(1+x2*(2.72^(-x3*1990)))-x1/(1+x2)-0.22 == 0,
x1/(1+x2*(2.72^(-x3*2002)))-x1/(1+x2)-0.31 == 0,
x1/(1+x2*(2.72^(-x3*2005)))-x1/(1+x2)-0.33 == 0},
{{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)}},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```
interface(displayprecision = 16)
with(RandomTools):
fsolve({x1/(1+x2*(2.72^(-x3*1990)))-x1/(1+x2)-0.22 = 0,
x1/(1+x2*(2.72^(-x3*2002)))-x1/(1+x2)-0.31 = 0,
x1/(1+x2*(2.72^(-x3*2005)))-x1/(1+x2)-0.33 = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1)})
```

The results for Case IV are shown in Table 4:

This is a very interesting problem. Although the two solutions reported by 1stOpt appear almost symmetric with respect to  $x_1$  and  $x_3$ , the symmetry is only superficial. One might expect that simply reversing the signs of  $x_1$  and  $x_3$  in either solution would yield the other solution, but this is not the case. While the magnitudes of the corresponding variables are remarkably close, the sign-reversed values do not satisfy the system of equations. In other words, the two solutions are not exact reflections of one another.

Another notable characteristic of this problem is that the parameter  $x_2$  takes on values that are either extremely small (on the order of 1E-58) or extremely large (on the order of 1E57). This wide range of values for  $x_2$  adds to the complexity of solving the system, as it can lead to numerical instability and convergence issues for many solvers. Despite having only three unknowns, this problem is particularly challenging for the '3M' software, which struggles to find the correct solution even when provided with initial values close to the optimal solution, such as  $x = [-0.5, 1e-55, -0.1]$ .



Table 4: Numerical solutions and residual errors for Case IV

|             | Parameter Values                                                                                                                                                                         | Residual values                                                                                                                                           |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1stOpt      | Solution 1:<br>x1: -0.466327667369193<br>x2: 3.18808468161272E-58<br>x3: -0.0664291202174226<br>Solution 2:<br>x1: 0.466327667302153<br>x2: 3.1366796282177E57<br>x3: 0.0664291202313091 | Solution 1:<br>6.16867668057353E-13<br>-4.93771690202038E-13<br>1.58900670399476E-12<br>Solution 2:<br>1.94289029309402E-16<br>-5.55111512312578E-17<br>0 |
| Matlab      | No correct solution                                                                                                                                                                      |                                                                                                                                                           |
| Mathematica | No correct solution                                                                                                                                                                      |                                                                                                                                                           |
| Maple       | No correct solution                                                                                                                                                                      |                                                                                                                                                           |

### 1.2.5 Case V: A four-equation system with four variables

$$x_1 + x_2 \cdot \exp\left(\frac{x_3}{a_i}\right) + x_4 \cdot a_i - b_i = 0, i = 1 \dots 4$$

Where: a=[1423.59,1430.57,1449.69,1481.98], b=[21.95,11.18,7.69,3]

Case V - Code Solution

1stOpt Code Solution:

```
Constant a=[1423.59,1430.57,1449.69,1481.98],b=[21.95,11.18,7.69,3];
Function For(a,b)(x1+x2*exp(x3/a)+x4*a-b=0);
```

Matlab Code Solution:

```
equations = @(x) [x(1)+x(2)*exp(x(3)/1423.59)+x(4)*1423.59-21.95-0;
x(1)+x(2)*exp(x(3)/1430.57)+x(4)*1430.57-11.18-0;
x(1)+x(2)*exp(x(3)/1449.69)+x(4)*1449.69-7.69-0;
x(1)+x(2)*exp(x(3)/1481.98)+x(4)*1481.98-3-0];
InitialGuess = 10*(2*rand(4,1)-1); %[1,1,1,1];
%InitialGuess = ones(4,1); %[1,1,1,1];
options = optimoptions('fsolve','Display','iter','MaxIterations',5000,'MaxFunEvals',5000);
[x,fval] = fsolve(equations,InitialGuess,options);
```

Mathematica Code Solution:

```
FindRoot[{x1+x2*Exp[x3/1423.59]+x4*1423.59-21.95-0 == 0,
x1+x2*Exp[x3/1430.57]+x4*1430.57-11.18-0 == 0,
x1+x2*Exp[x3/1449.69]+x4*1449.69-7.69-0 == 0,
x1+x2*Exp[x3/1481.98]+x4*1481.98-3-0 == 0},
{{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)},
{x4, 10*(2*RandomReal[]-1)}},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```
interface(displayprecision = 16)
with(RandomTools):
fsolve({x1+x2*exp(x3/1423.59)+x4*1423.59-21.95-0 = 0,
x1+x2*exp(x3/1430.57)+x4*1430.57-11.18-0 = 0,
x1+x2*exp(x3/1449.69)+x4*1449.69-7.69-0 = 0,
x1+x2*exp(x3/1481.98)+x4*1481.98-3-0 = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1),
x4=10*(2*Generate(float)-1)})
```

The results for Case V are shown in Table 5:

This is a highly distinctive system of equations. According to the reference solution obtained by 1stOpt, variables  $x_2$  attains a magnitude of  $10^{-239}$ , yet it cannot be treated as zero. The verification and comparison results presented in Table 6 clearly illustrate an extreme sensitivity to perturbations, where a tiny change in the value of  $x_2$  can lead to a significant increase in the residual errors of the equations.

For the '3M' software, it might be too challenging to find the correct solution when the initial values are randomly generated within the range of  $[-10, 10]$ . However, even when the initial values are set to be extremely close to the optimal solution, such as  $x = [200, 1e-235, 800000, -0.15]$  (although it is nearly impossible to guess such initial values in the first place), a correct solution still cannot be achieved. The results suggest that the system poses severe numerical challenges and exposes limi understandable that a correct solution cannot be obtained with random initial values within the range  $[-10, 10]$ . However, even when given initial values extremely close to the optimal values, such as  $x = [200, 1e-235, 800000, -0.15]$  (although it is nearly impossible to guess such initial values manually), a correct solution still cannot be achieved. This behavior highlights the considerable numerical difficulty posed by this system.

Table 5: Numerical solutions and residual errors for Case V

|             | Parameter Values                                                                                                                                                | Residual values                                                                                  |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 1stOpt      | x1: 218.228110910674<br>x2: 9.10236261977445E-239<br>x3: 783626.245975248<br>x4: -0.145230104937087                                                             | -1.94333438230387E-12<br>-1.01607611213694E-12<br>-7.13207271019201E-13<br>-7.95807864051312E-13 |
| Matlab      | Can only provide approximate real solutions, such as:<br>x1 = 7.59305893673008<br>x2 = -7.95679582994413<br>x3 = -0.995505844279068<br>x4 = 0.00775403955282771 | -11.2696015433881<br>-0.445505476725685<br>3.19267878262024<br>8.13293774974168                  |
| Mathematica | No correct solution                                                                                                                                             |                                                                                                  |
| Maple       | No correct solution                                                                                                                                             |                                                                                                  |

Table 6: Verification of the impact of  $x_2$  values on equation errors for Case V

| $x_2$ value           | Residual values                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------|
| 9.10236262007253E-239 | 2.43360886997834E-12<br>1.49213974509621E-13<br>8.26005930321116E-14<br>1.13686837721616E-13 |
| 9.10236261977445E-238 | 94.2301275904448<br>6.42348278256942<br>0.00467924014236321<br>3.59085277068516E-8           |
| 9.10236261977445E-240 | -9.42301275904178<br>-0.642348278256769<br>-0.000467924014148302<br>-3.59074192601838E-9     |
| 0                     | -10.4700141767132<br>-0.7137203091741<br>-0.00051991557120612<br>-3.98969746129296E-9        |

### 1.2.6 Case VI: A four-equation system with four variables

$$\begin{cases} 1 - \exp(-(x_1 + x_2 \cdot 915) \cdot 1200^{x_3+x_4 \cdot 915}) = \frac{1.600}{100} \\ 1 - \exp(-(x_1 + x_2 \cdot 909) \cdot 1212^{x_3+x_4 \cdot 909}) = \frac{3.540}{100} \\ 1 - \exp(-(x_1 + x_2 \cdot 898) \cdot 1234^{x_3+x_4 \cdot 898}) = \frac{5.430}{100} \\ 1 - \exp(-(x_1 + x_2 \cdot 895) \cdot 1240^{x_3+x_4 \cdot 895}) = \frac{5.683}{100} \end{cases}$$

Case VI - Code Solution

1stOpt Code Solution:

```
Function
1-exp(-(x1+x2*915)*1200^(x3+x4*915))=1.600/100;
1-exp(-(x1+x2*909)*1212^(x3+x4*909))=3.540/100;
1-exp(-(x1+x2*898)*1234^(x3+x4*898))=5.430/100;
1-exp(-(x1+x2*895)*1240^(x3+x4*895))=5.683/100;
```

Matlab Code Solution:

```
equations = @(x) [1-exp(0-(x(1)+x(2)*915)*1200^(x(3)+x(4)*915))-(1.600/100);
1-exp(0-(x(1)+x(2)*909)*1212^(x(3)+x(4)*909))-(3.540/100);
1-exp(0-(x(1)+x(2)*898)*1234^(x(3)+x(4)*898))-(5.430/100);
1-exp(0-(x(1)+x(2)*895)*1240^(x(3)+x(4)*895))-(5.683/100)];
InitialGuess = 10*(2*rand(4,1)-1); %[1,1,1,1];
%InitialGuess = ones(4,1); %[1,1,1,1];
options = optimoptions('fsolve','Display','iter','MaxIterations',5000,'MaxFunEvals',5000);
[x,fval] = fsolve(equations,InitialGuess,options);
```

Mathematica Code Solution:

```
FindRoot[{1-Exp[0-(x1+x2*915)*1200^(x3+x4*915)]-(1.600/100) == 0,
1-Exp[0-(x1+x2*909)*1212^(x3+x4*909)]-(3.540/100) == 0,
1-Exp[0-(x1+x2*898)*1234^(x3+x4*898)]-(5.430/100) == 0,
1-Exp[0-(x1+x2*895)*1240^(x3+x4*895)]-(5.683/100) == 0},
{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)},
{x4, 10*(2*RandomReal[]-1)},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```
interface(displayprecision = 16)
with(RandomTools):
fsolve({1-exp(0-(x1+x2*915)*1200^(x3+x4*915))-(1.600/100) = 0,
1-exp(0-(x1+x2*909)*1212^(x3+x4*909))-(3.540/100) = 0,
1-exp(0-(x1+x2*898)*1234^(x3+x4*898))-(5.430/100) = 0,
1-exp(0-(x1+x2*895)*1240^(x3+x4*895))-(5.683/100) = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1),
x4=10*(2*Generate(float)-1)})
```

The results for Case VI are shown in Table 7:

None of the three "3M" software succeeded in obtaining the correct solution under the tested conditions, whereas 1stOpt consistently converged to the correct solution in all runs.

Table 7: Numerical solutions and residual errors for Case VI

|             | Parameter Values                                                                                                                                                                   | Residual values                                                                               |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 1stOpt      | x1: 180269.749976749<br>x2: -196.244818619846<br>x3: -4.99648566803081<br>x4: 0.00381339657343384                                                                                  | -2.08166817117217E-16<br>3.19189119579733E-16<br>1.13103970633688E-15<br>9.36750677027476E-16 |
| Matlab      | No correct solution                                                                                                                                                                |                                                                                               |
| Mathematica | Either no solution is obtained, or only an approximate real solution is returned, such as:<br>4.053004656357142<br>2.842659487302639<br>-2.402054449041441<br>-0.07449075855549998 | -0.016<br>-0.0354<br>-0.0543<br>-0.05683                                                      |
| Maple       | No correct solution                                                                                                                                                                |                                                                                               |

### 1.2.7 Case VII: A four-equation system with four variables

$$\begin{cases} 6250 \cdot \pi \cdot x_1 + 2500 \cdot \pi \cdot x_1 \cdot (15 - x_2) + 10.5 - x_3 = 0 \\ -169169 \cdot x_2 + 168971 \cdot x_2 - x_4 \cdot x_3 - 1171.7 = 0 \\ 1.03 \cdot x_2 + 1.84 \cdot x_2 \cdot x_4 \cdot x_3 + 2500 \cdot x_4 \cdot x_3 - x_3 = 0 \\ 6166 - 282 \cdot x_2 - x_1 \cdot (20 - x_2) = 0 \end{cases}$$

Case VII - Code Solution

1stOpt Code Solution:

```
Hardness = 2; //1,2,...5
Function
6250*pi*x1+2500*pi*x1*(15-x2)+10.5-x3=0;
-169169*x2+168971*x2-x4*x3-1171.7=0;
1.03*x2+1.84*x2*x4*x3+2500*x4*x3-x3=0;
6166-282*x2-x1*(20-x2)=0;
```

Matlab Code Solution:

```
equations = @(x) [1-exp(0-(x(1)+x(2)*915)*1200^(x(3)+x(4)*915))-(1.600/100);
1-exp(0-(x(1)+x(2)*909)*1212^(x(3)+x(4)*909))-(3.540/100);
1-exp(0-(x(1)+x(2)*898)*1234^(x(3)+x(4)*898))-(5.430/100);
1-exp(0-(x(1)+x(2)*895)*1240^(x(3)+x(4)*895))-(5.683/100)];
InitialGuess = 10*(2*rand(4,1)-1); %[1,1,1,1];
%InitialGuess = ones(4,1); %[1,1,1,1];
options = optimoptions('fsolve','Display','iter','MaxIterations',5000,'MaxFunEvals',5000);
[x,fval] = fsolve(equations,InitialGuess,options);
```

Mathematica Code Solution:

```
FindRoot[{1-Exp[0-(x1+x2*915)*1200^(x3+x4*915)]-(1.600/100) == 0,
1-Exp[0-(x1+x2*909)*1212^(x3+x4*909)]-(3.540/100) == 0,
1-Exp[0-(x1+x2*898)*1234^(x3+x4*898)]-(5.430/100) == 0,
1-Exp[0-(x1+x2*895)*1240^(x3+x4*895)]-(5.683/100) == 0},
{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)},
{x4, 10*(2*RandomReal[]-1)}},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```
interface(displayprecision = 16)
with(RandomTools):
fsolve({1-exp(0-(x1+x2*915)*1200^(x3+x4*915))-(1.600/100) = 0,
1-exp(0-(x1+x2*909)*1212^(x3+x4*909))-(3.540/100) = 0,
1-exp(0-(x1+x2*898)*1234^(x3+x4*898))-(5.430/100) = 0,
1-exp(0-(x1+x2*895)*1240^(x3+x4*895))-(5.683/100) = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1),
x4=10*(2*Generate(float)-1)})
```

The results for Case VII are shown in Table 8:

In all runs, 1stOpt reliably converges to one of the valid solutions, while the three "3M" software fail to find any correct solution under the tested conditions.

Table 8: Numerical solutions and residual errors for Case VII

|             | Parameter Values                                                                                                                                                                                                                    | Residual values                                                                                                                                                                                                            |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1stOpt      | Solution 1:<br>x1: 213.02505351492<br>x2: 27.6259573483508<br>x3: -16941676.6706643<br>x4: 0.000392029648781691<br>Solution 2:<br>x1: 942.493330252235<br>x2: 19.2036255690895<br>x3: -12610780.1619252<br>x4: 0.000394425864126739 | Solution 1:<br>-8.19563865661621E-8<br>-1.20508047984913E-11<br>1.11758708953857E-8<br>9.09494701772928E-13<br>Solution 2:<br>-1.08033418655396E-7<br>-3.20596882374957E-11<br>1.49011611938477E-8<br>1.94972926692571E-10 |
| Matlab      | No correct solution                                                                                                                                                                                                                 |                                                                                                                                                                                                                            |
| Mathematica | No correct solution                                                                                                                                                                                                                 |                                                                                                                                                                                                                            |
| Maple       | No correct solution                                                                                                                                                                                                                 |                                                                                                                                                                                                                            |

### 1.2.8 Case VIII: A five-equation system with five variables

$$\begin{cases} -x_1 - x_2 \cdot x_3 - x_4 \cdot x_5 + 12.04 = 0 \\ x_2 \cdot x_3^2 - x_4 \cdot x_5^2 - 20.70 = 0 \\ -1.30 \cdot x_1 - x_2 \cdot \ln(1 + 1.30 \cdot x_3) - x_4 \cdot (\exp(1.30 \cdot x_5) - 1) - 4.66 = 0 \\ -x_1 - \frac{x_2 \cdot x_3}{1 + 1.30 \cdot x_3} - x_4 \cdot x_5 \cdot \exp(1.30 \cdot x_5) - 52.57 = 0 \\ \frac{x_2 \cdot x_3^2}{(1 + 1.30 \cdot x_3)^2} - x_4 \cdot x_5^2 \cdot \exp(1.30 \cdot x_5) - 109.83 = 0 \end{cases}$$

Case VIII - Code Solution

1stOpt Code Solution:

```
Function -x1-x2*x3-x4*x5+12.04=0;
x2*x3^2-x4*x5^2-20.70=0;
-1.30*x1-x2*ln(1+1.30*x3)-x4*(exp(1.30*x5)-1)-4.66=0;
-x1-x2*x3/(1+1.30*x3)-x4*x5*exp(1.30*x5)-52.57=0;
x2*x3^2/(1+1.30*x3)^2-x4*x5^2*exp(1.30*x5)-109.83=0;
```

Matlab Code Solution:

```
equations = @(x) [-x(1)-x(2)*x(3)-x(4)*x(5)+12.04-0;
x(2)*x(3)^2-x(4)*x(5)^2-20.70-0;
-1.30*x(1)-x(2)*log(1+1.30*x(3))-x(4)*(exp(1.30*x(5))-1)-4.66-0;
-x(1)-x(2)*x(3)/(1+1.30*x(3))-x(4)*x(5)*exp(1.30*x(5))-52.57-0;
x(2)*x(3)^2/(1+1.30*x(3))^2-x(4)*x(5)^2*exp(1.30*x(5))-109.83-0];
InitialGuess = 10*(2*rand(5,1)-1); % [1,1,1,1,1];
options = optimoptions('fsolve','Display','iter','MaxFunEvals',5000);
[x,fval] = fsolve(equations,InitialGuess,options);
```

Mathematica Code Solution:

```
FindRoot[{-x1-x2*x3-x4*x5+12.04-0 == 0,
x2*x3^2-x4*x5^2-20.70-0 == 0,
-1.30*x1-x2*Log[1+1.30*x3]-x4*(Exp[1.30*x5]-1)-4.66-0 == 0,
-x1-x2*x3/(1+1.30*x3)-x4*x5*Exp[1.30*x5]-52.57-0 == 0,
x2*x3^2/(1+1.30*x3)^2-x4*x5^2*Exp[1.30*x5]-109.83-0 == 0},
{{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)},
{x4, 10*(2*RandomReal[]-1)},
{x5, 10*(2*RandomReal[]-1)}},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```
interface(displayprecision = 16)
with(RandomTools):
fsolve({-x1-x2*x3-x4*x5+12.04-0 = 0,
x2*x3^2-x4*x5^2-20.70-0 = 0,
-1.30*x1-x2*log(1+1.30*x3)-x4*(exp(1.30*x5)-1)-4.66-0 = 0,
-x1-x2*x3/(1+1.30*x3)-x4*x5*exp(1.30*x5)-52.57-0 = 0,
x2*x3^2/(1+1.30*x3)^2-x4*x5^2*exp(1.30*x5)-109.83-0 = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1),
x4=10*(2*Generate(float)-1), x5=10*(2*Generate(float)-1)})
```

The results for Case VIII are shown in Table 9:

1stOpt and Maple can consistently obtain the correct solution, while Matlab and Mathematica fail to find the correct result.



Table 9: Numerical solutions and residual errors for Case VIII

|             | Parameter Values                                                                                                                                                                                                                                 | Residual values                                                                                                      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 1stOpt      | x1: -499.250341459814<br>x2: 1124.66857461431<br>x3: 1.12507902890424<br>x4: 405.29540935769<br>x5: -1.86049648946527                                                                                                                            | 6.28830321147689E-12<br>-9.27613541534811E-12<br>3.94706489714736E-12<br>2.97717406283482E-12<br>-7.105427357601E-14 |
| Matlab      | Either there is no solution, or only an approximate solution far from the correct answer is obtained, such as:<br>x1 = 10199.7734319073<br>x2 = -4309990.72505758<br>x3 = 0.00236045541019449<br>x4 = -12.8403157269356<br>x5 = 1.35952926013614 | 3.26427788309106<br>-20.9811798028529<br>3.33381993445832<br>-7.7065663411042<br>5.27207755255068                    |
| Mathematica | No real solution                                                                                                                                                                                                                                 |                                                                                                                      |
| Maple       | x1 = -499.2503415000000000<br>x2 = 1124.6685750000000000<br>x3 = 1.1250790290000000<br>x4 = 405.2954094000000000<br>x5 = -1.8604964890000000                                                                                                     | -6.11296023578234E-7<br>1.28575429769739E-6<br>-3.35487047919969E-7<br>-1.22951327341525E-7<br>7.06111364934259E-8   |

### 1.2.9 Case IX: A five-equation system with five variables

$$\begin{cases} 0.0448 - x_1 \cdot \left( \frac{x_2 - 0.0633}{x_3 - 0.0633} \right)^{x_5} = 0 \\ 0.0471 - x_1 \cdot \left( \frac{x_2 - 0.0333}{x_4 - 0.0333} \right)^{x_5} = 0 \\ 0.4714 - x_1 \cdot \left( \frac{x_2 + 0.0633}{x_3 + 0.0633} \right)^{x_5} = 0 \\ 0.5715 - x_1 \cdot \left( \frac{x_2 + 0.7}{0.5946 \cdot x_4 + 0.5 \cdot x_3 + 0.7} \right)^{x_5} = 0 \\ 0.5657 - x_1 \cdot \left( \frac{x_2 + 0.8}{x_4 + 0.8} \right)^{x_5} = 0 \end{cases}$$

Case IX - Code Solution

1stOpt Code Solution:

```
Function 0.0448-x1*((x2-0.0633)/(x3-0.0633))^x5=0;
        0.0471-x1*((x2-0.0333)/(x4-0.0333))^x5=0;
        0.4714-x1*((x2+0.3333)/(x3+0.3333))^x5=0;
        0.5715-x1*((x2+0.7)/(0.5946*x4+0.5*x3+0.7))^x5=0;
        0.5657-x1*((x2+0.8)/(x4+0.8))^x5=0;
```

Matlab Code Solution:

```
equations = @(param) [0.0448-param(1)*((param(2)-0.0633)/(param(3)-0.0633))^param(5)-0;
0.0471-param(1)*((param(2)-0.0333)/(param(4)-0.0333))^param(5)-0;
0.4714-param(1)*((param(2)+0.3333)/(param(3)+0.3333))^param(5)-0;
0.5715-param(1)*((param(2)+0.7)/(0.5946*param(4)+0.5*param(3)+0.7))^param(5)-0;
0.5657-param(1)*((param(2)+0.8)/(param(4)+0.8))^param(5)-0];
InitialGuess = 10*(2*rand(5,1)-1); %[1,1,1,1,1];
options = optimoptions('fsolve','Display','iter','MaxIterations',5000,'MaxFunEvals',5000);
[param,fval] = fsolve(equations,InitialGuess,options);
x1 = param(1);
x2 = param(2);
x3 = param(3);
x4 = param(4);
x5 = param(5);
```

Mathematica Code Solution:

```
FindRoot[{0.0448-x1*((x2-0.0633)/(x3-0.0633))^x5-0 == 0,
0.0471-x1*((x2-0.0333)/(x4-0.0333))^x5-0 == 0,
0.4714-x1*((x2+0.3333)/(x3+0.3333))^x5-0 == 0,
0.5715-x1*((x2+0.7)/(0.5946*x4+0.5*x3+0.7))^x5-0 == 0,
0.5657-x1*((x2+0.8)/(x4+0.8))^x5-0 == 0},
{{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)},
{x5, 10*(2*RandomReal[]-1)},
{x4, 10*(2*RandomReal[]-1)}},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```
interface(displayprecision = 16)
with(RandomTools):
fsolve({0.0448-x1*((x2-0.0633)/(x3-0.0633))^x5-0 = 0,
0.0471-x1*((x2-0.0333)/(x4-0.0333))^x5-0 = 0,
0.4714-x1*((x2+0.3333)/(x3+0.3333))^x5-0 = 0,
```

```

0.5715-x1*((x2+0.7)/(0.5946*x4+0.5*x3+0.7))^x5-0 = 0,
0.5657-x1*((x2+0.8)/(x4+0.8))^x5-0 = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1),
x4=10*(2*Generate(float)-1), x5=10*(2*Generate(float)-1))}

```

The results for Case IX are shown in Table 10:

Table 10: Numerical solutions and residual errors for Case IX

|             | Parameter Values                                                                                                                                                                                                               | Residual values                                                                                                       |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 1stOpt      | x1: 0.754428363386432<br>x2: 0.21999193531669<br>x3: 0.0988142336799259<br>x4: 0.0767421031370608<br>x5: -1.90235171609445                                                                                                     | 4.26721158408583E-13<br>5.63757374116847E-13<br>6.45983266878147E-13<br>-1.3935519405095E-12<br>-7.54396545232794E-13 |
| Matlab      | Either there is a complex solution, or only an approximate solution is obtained, such as:<br>x1 = -6.23003017369515<br>x2 = 12.6627724935181<br>x3 = 2.20258455438707<br>x4 = 0.793068719981685<br>x5 = -10.3184329249217      | 0.04480007053944515<br>0.0471000000015803<br>0.47140029626296287<br>0.5715000718216434<br>0.5657000016995529          |
| Mathematica | Either there is a complex solution, or only an approximate solution is obtained, such as:<br>x1 = 3.552883986619389<br>x2 = 2.960818240360410<br>x3 = 0.1623061162807060<br>x4 = 0.8826537191916343<br>x5 = -2.161584660196593 | 0.0423961311745995<br>-0.197762126230464<br>0.412181687986102<br>0.188706129341528<br>-0.0588487157903473             |
| Maple       | Execution time is excessively long, and no correct solution is reported.                                                                                                                                                       |                                                                                                                       |

In our experiments, Matlab and Mathematica returned complex solutions in approximately 70% of the runs, and in the remaining 30% produced real solutions that deviated substantially from the correct answer. Maple did not obtain a solution within a reasonable time frame. In contrast, 1stOpt consistently converged to the correct solution in all runs.

### 1.2.10 Case X: A six-equation system with six variables

$$\begin{cases} x_1^2 + x_3^2 - 1 = 0 \\ x_2^2 + x_4^2 - 1 = 0 \\ x_5 \cdot x_3^3 + x_6 \cdot x_4^3 - 100 = 0 \\ x_5 \cdot x_1^{x_3} + x_6 \cdot x_2^3 + 20 = 0 \\ x_5 \cdot x_1 \cdot x_3^{x_2} + x_6 \cdot x_2 \cdot x_4^2 - 5.5 = 0 \\ x_5 \cdot x_3 \cdot x_1^2 + x_6 \cdot x_4 \cdot x_2^2 + c = 0 \end{cases}$$

where c can be 30, 40, 50, 60.

Case X - Code Solution

1stOpt Code Solution:

```
Parameter x(6);
Constant c4=30;//40,50,60
Function x1^2+x3^2-1=0;
      x2^2+x4^2-1=0;
      x5*x3^3+x6*x4^3-100=0;
      x5*x1^x3+x6*x2^3+20=0;
      x5*x1*x3^x2+x6*x2*x4^2-5.5=0;
      x5*x3*x1^2+x6*x4*x2^2+c4=0;
```

Matlab Code Solution:

```
equations = @(x) [x(1)^2+x(3)^2-1-0;
x(2)^2+x(4)^2-1-0;
x(5)*x(3)^3+x(6)*x(4)^3-100-0;
x(5)*x(1)^x(3)+x(6)*x(2)^3+20-0;
x(5)*x(1)*x(3)^x(2)+x(6)*x(2)*x(4)^2-5.5-0;
x(5)*x(3)*x(1)^2+x(6)*x(4)*x(2)^2+30-0];
InitialGuess = 10*(2*rand(6,1)-1);
options = optimoptions('fsolve','Display','iter','MaxFunEvals',5000);
[x,fval] = fsolve(equations,InitialGuess,options);
```

Mathematica Code Solution:

```
FindRoot[{x1^2+x3^2-1-0 == 0,
x2^2+x4^2-1-0 == 0,
x5*x3^3+x6*x4^3-100-0 == 0,
x5*x1^x3+x6*x2^3+20-0 == 0,
x5*x1*x3^x2+x6*x2*x4^2-5.5-0 == 0,
x5*x3*x1^2+x6*x4*x2^2+30-0 == 0},
{{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)},
{x4, 10*(2*RandomReal[]-1)},
{x5, 10*(2*RandomReal[]-1)},
{x6, 10*(2*RandomReal[]-1)}},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```
interface(displayprecision = 16)
with(RandomTools):
fsolve({x1^2+x3^2-1-0 = 0,
x2^2+x4^2-1-0 = 0,
x5*x3^3+x6*x4^3-100-0 = 0,
x5*x1^x3+x6*x2^3+20-0 = 0,
x5*x1*x3^x2+x6*x2*x4^2-5.5-0 = 0,
```

```
x5*x3*x1^2+x6*x4*x2^2+30-0 = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1),
x4=10*(2*Generate(float)-1), x5=10*(2*Generate(float)-1), x6=10*(2*Generate(float)-1)}
```

The results for Case X ( $c = 30$ ) are shown in Table 11:

Table 11: Numerical solutions and residual errors for Case X ( $c = 30$ )

|             | Parameter Values                                                                                                                                                                                                                                                                                                                     | Residual values                                                                                                                                                                                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1stOpt      | Solution 1:<br>x1: 0.240184968662544<br>x2: -0.815496094330282<br>x3: 0.970727140255448<br>x4: -0.578762576651357<br>x5: 129.845222134722<br>x6: 96.8346666139437<br>Solution 2:<br>x1: 0.251817492626888<br>x2: 0.81977727798232<br>x3: 0.967774741562773<br>x4: 0.572682472669906<br>x5: 130.799940226334<br>x6: -98.8067228339781 | Solution 1:<br>-5.25135490647699E-14<br>6.21724893790088E-14<br>1.4210854715202E-14<br>-6.60804744256893E-13<br>-3.01980662698043E-13<br>7.92255150372512E-13<br>Solution 2:<br>-2.15161222172355E-13<br>-5.80424597274032E-13<br>-6.11066752753686E-13<br>-3.12638803734444E-13<br>1.80833126250945E-12<br>1.13686837721616E-13 |
| Matlab      | Either a complex solution is obtained, or only an approximate real solution is returned, such as:<br>x1 = 1.78781411286833<br>x2 = -0.27333847371648<br>x3 = 1.23289812400307<br>x4 = -1.58268087994283<br>x5 = -8.65317451075732<br>x6 = -29.3003518844299                                                                          | 3.71631708634146<br>1.57959268895027<br>-0.0574863091199802<br>2.88655577161781<br>-0.0484310407171815<br>-0.634733546377031                                                                                                                                                                                                     |
| Mathematica | only complex solutions are reported                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                  |
| Maple       | x1 = 0.2518174926<br>x2 = 0.8197772780<br>x3 = 0.9677747416<br>x4 = 0.5726824727<br>x5 = 130.7999402000<br>x6 = -98.8067228300                                                                                                                                                                                                       | 5.8297811023067E-11<br>6.28754825982014E-11<br>-1.23667689422291E-8<br>-1.35894069330789E-8<br>-1.11803579727621E-8<br>-5.12887510240034E-9                                                                                                                                                                                      |

The results for Case X ( $c = 40$ ) are shown in Table 12:

The results for Case X ( $c = 50$ ) are shown in Table 13:

The results for Case X ( $c = 60$ ) are shown in Table 14:

For all tested values of  $c$ , MATLAB consistently returns either complex solutions or real solutions that deviate significantly from the correct ones. Mathematics is able to obtain one of the correct solution in roughly 10% of the runs when  $c = 40$  or  $c = 60$ , but in most cases it still produces complex or incorrect solutions. Maple can successfully compute one of the correct solution when  $c < 50$ , but fails to produce an accurate result once  $c \geq 50$ , with all returned solutions being inaccurate and incorrect. In contrast, 1stOpt converges to one of the two correct solutions with a 100% success rate across all tested values of  $c$  in each run, and is the only software capable of reliably finding both solutions across all tested cases.

Table 12: Numerical solutions and residual errors for Case X ( $c = 40$ )

|             | Parameter Values                                                                                                                                                                                                                                                                                                                    | Residual values                                                                                                                                                                                                                                                                                                                  |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1stOpt      | Solution 1:<br>x1: 0.318434845365605<br>x2: 0.796377176974262<br>x3: 0.947944750107746<br>x4: 0.604800291000486<br>x5: 154.535909765989<br>x6: -143.008370949549<br>Solution 2:<br>x1: 0.29672818202995<br>x2: -0.788168611728715<br>x3: 0.954961981436569<br>x4: -0.61545937273343<br>x5: 151.767945391027<br>x6: 137.998739822563 | Solution 1:<br>-1.46105350040671E-13<br>-2.32258656751583E-13<br>-2.55795384873636E-13<br>3.5527136788005E-14<br>-7.74491581978509E-13<br>-1.13686837721616E-13<br>Solution 2:<br>5.7065463465733E-14<br>-2.02948768901479E-13<br>-2.8421709430404E-13<br>-1.63424829224823E-13<br>-1.20792265079217E-13<br>-4.9737991503207E-13 |
| Matlab      | Either a complex solution is obtained, or only an approximate real solution is returned, such as:<br>x1 = 0.0590606508881057<br>x2 = 0.0149030930797581<br>x3 = 2.01186063784314<br>x4 = 2.0070223796826<br>x5 = -6025.83624320978<br>x6 = 6081.88954106651                                                                         | 3.05107138658593<br>3.02836093473015<br>0.000778685738623608<br>-0.30533885744925<br>-0.0105688570369011<br>0.423620262360146                                                                                                                                                                                                    |
| Mathematica | x1 = 0.3184348453656811<br>x2 = 0.7963771769743352<br>x3 = 0.9479447501077975<br>x4 = 0.6048002910005819<br>x5 = 154.5359097659733<br>x6 = -143.0083709495165                                                                                                                                                                       | 6.66133814775094E-16<br>-2.22044604925031E-16<br>0<br>1.49213974509621E-13<br>5.6843418860808E-14<br>1.63424829224823E-13                                                                                                                                                                                                        |
| Maple       | x1 = 0.3184348454<br>x2 = 0.7963771770<br>x3 = 0.9479447501<br>x4 = 0.6048002910<br>x5 = 154.5359098000<br>x6 = -143.0083709000                                                                                                                                                                                                     | 7.0734529344918E-12<br>4.01738642352711E-11<br>3.67819268376479E-8<br>3.53286964127619E-8<br>2.82543624052778E-8<br>2.18608064983528E-8                                                                                                                                                                                          |

Table 13: Numerical solutions and residual errors for Case X ( $c = 50$ )

|             | Parameter Values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Residual values                                                                                                                                                                                                                                                                                                              |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1stOpt      | Solution 1:<br>x1: 0.337889417595571<br>x2: -0.771116171737518<br>x3: 0.941185816657442<br>x4: -0.636694471222222<br>x5: 176.3205266392<br>x6: 182.113060658168<br>Solution 2:<br>x1: 0.369498005183815<br>x2: 0.783455018890168<br>x3: 0.929231523445886<br>x4: 0.621448496156864<br>x5: 181.940506071967<br>x6: -191.592472055481                                                                                                                                                                                                               | Solution 1:<br>2.10276240864005E-13<br>7.01660951563099E-14<br>-2.70006239588838E-13<br>-2.06057393370429E-13<br>4.47641923528863E-13<br>4.2632564145606E-14<br>Solution 2:<br>3.80806497446429E-13<br>-1.78634884662188E-13<br>4.2632564145606E-14<br>-2.8421709430404E-14<br>-1.84741111297626E-13<br>2.98427949019242E-13 |
| Matlab      | Either a complex solution is obtained, or only an approximate real solution is returned, such as:<br>x1 = 2.28610654556004<br>x2 = 0.365799039040494<br>x3 = 0.982731578881764<br>x4 = 1.30746667881677<br>x5 = -11.1966330866053<br>x6 = 49.4896479910809                                                                                                                                                                                                                                                                                        | 5.1920444937839<br>0.843278053179104<br>-0.0135264209355341<br>-2.81143452066038<br>0.0128596831320991<br>1.15196747464032                                                                                                                                                                                                   |
| Mathematica | Return complex solutions only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                              |
| Maple       | Return inaccurate and incorrect solutions, such as:<br>x1 = -4.0165291200000000<br>x2 = 1.4818083500000000<br>x3 = 9.4994309900000000<br>x4 = -3.8886403260000000<br>x5 = -0.2980721020000000<br>x6 = 6.6220511700000000<br>This solution leads to a computational error in the fourth equation when evaluating the term $x_1^{x_3}$ , because $x_1$ is negative while $x_3$ is a positive non-integer, making the expression mathematically impossible to compute. Moreover, the residuals of the other equations also fail to converge to zero. | 105.37169530558<br>16.3172795711231<br>-744.904605204945<br>NAN<br>176.527567484613<br>-52.2218579069458                                                                                                                                                                                                                     |

Table 14: Numerical solutions and residual errors for Case X ( $c = 60$ )

|             | Parameter Values                                                                                                                                                                                                                                                                                                                      | Residual values                                                                                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1stOpt      | Solution 1:<br>x1: 0.367956965401267<br>x2: -0.759675764854691<br>x3: 0.929842820917984<br>x4: -0.650302031591813<br>x5: 202.286679896916<br>x6: 227.732361079941<br>Solution 2:<br>x1: 0.408164232727109<br>x2: 0.775865720168232<br>x3: 0.912908516293949<br>x4: 0.630898077559241<br>x5: 211.543983334219<br>x6: -242.702124800984 | Solution 1:<br>2.30926389122033E-14<br>-8.88178419700125E-16<br>0<br>4.2632564145606E-14<br>2.70006239588838E-13<br>1.77635683940025E-13<br>Solution 2:<br>-2.71116462613463E-13<br>1.15463194561016E-13<br>-1.13686837721616E-13<br>-2.8421709430404E-13<br>-1.13686837721616E-13<br>6.3948846218409E-14 |
| Matlab      | Either a complex solution is obtained, or an inaccurate and incorrect solutions is returned, such as:<br>x1 = -9.34798358938944<br>x2 = 1.2239958541932<br>x3 = 7.6373300090362<br>x4 = 3.38350609068788<br>x5 = -6.19133465640092<br>x6 = -2.6216690787221                                                                           | 144.713606854419<br>11.9462793168041<br>-2959.64300551595<br>NAN<br>654.741939490097<br>-4085.30275675849                                                                                                                                                                                                 |
| Mathematica | 0.3679569654012514<br>-0.7596757648546925<br>0.9298428209179777<br>-0.6503020315918118<br>202.2866798969178<br>227.7323610799354                                                                                                                                                                                                      | 1.199678896029<br>0.186238604825<br>-8.58661198899171<br>3.43076838937441<br>-0.112474515079839<br>6.08868927844935                                                                                                                                                                                       |
| Maple       | Return inaccurate and incorrect solutions, such as:<br>x1 = -6.0652071700000000<br>x2 = -9.8290272450000000<br>x3 = -6.8848669660000000<br>x4 = 4.7493762900000000<br>x5 = -0.4124742360000000<br>x6 = 5.9163608700000000                                                                                                             | -6.0652071700000000<br>-9.8290272450000000<br>-6.8848669660000000<br>4.7493762900000000<br>-0.4124742360000000<br>5.9163608700000000                                                                                                                                                                      |



### 1.2.11 Case XI: A six-equation system with six variables

$$\begin{cases} 158.49 \cdot (1.47 + x_4) = 1.47 \cdot \left( \frac{1.74 + x_2 \cdot x_4}{0.23} \right)^{1.19} \\ (x_1 - 26.3 \cdot x_4) \cdot (1.47 + x_4) = x_4 \cdot \left( \frac{1.74 + x_2 \cdot x_4}{x_2 \cdot x_3} \right)^{x_2} \\ 173.78 \cdot (1.76 + x_5) = 1.76 \cdot \left( \frac{2.09 + x_2 \cdot x_5}{0.23} \right)^{1.19} \\ (x_1 - 13.2 \cdot x_5) \cdot (1.76 + x_5) = x_5 \cdot \left( \frac{2.09 + x_2 \cdot x_5}{x_2 \cdot x_3} \right)^{x_2} \\ 186.21 \cdot (2.01 + x_6) = 2.01 \cdot \left( \frac{2.39 + x_2 \cdot x_6}{0.23} \right)^{1.19} \\ (x_1 - 5.4 \cdot x_6) \cdot (2.01 + x_6) = x_6 \cdot \left( \frac{2.39 + x_2 \cdot x_6}{x_2 \cdot x_3} \right)^{x_2} \end{cases}$$

Case XI - Code Solution

1stOpt Code Solution:

```
Function
158.49*(1.47+x4)=1.47*((1.74+x2*x4)/0.23)^1.19;
(x1-26.3*x4)*(1.47+x4)=x4*((1.74+x2*x4)/(x2*x3))^x2;
173.78*(1.76+x5)=1.76*((2.09+x2*x5)/0.23)^1.19;
(x1-13.2*x5)*(1.76+x5)=x5*((2.09+x2*x5)/(x2*x3))^x2;
186.21*(2.01+x6)=2.01*((2.39+x2*x6)/0.23)^1.19;
(x1-5.4*x6)*(2.01+x6)=x6*((2.39+x2*x6)/(x2*x3))^x2;
```

Matlab Code Solution:

```
equations = @(x) [158.49*(1.47+x(4))-(1.47*((1.74+x(2)*x(4))/0.23)^1.19);
(x(1)-26.3*x(4))*(1.47+x(4))-(x(4)*((1.74+x(2)*x(4))/(x(2)*x(3)))^x(2));
173.78*(1.76+x(5))-(1.76*((2.09+x(2)*x(5))/0.23)^1.19);
(x(1)-13.2*x(5))*(1.76+x(5))-(x(5)*((2.09+x(2)*x(5))/(x(2)*x(3)))^x(2));
186.21*(2.01+x(6))-(2.01*((2.39+x(2)*x(6))/0.23)^1.19);
(x(1)-5.4*x(6))*(2.01+x(6))-(x(6)*((2.39+x(2)*x(6))/(x(2)*x(3)))^x(2))];
InitialGuess = 10*(2*rand(6,1)-1); % [1,1,1,1,1,1];
%InitialGuess = ones(6,1); % [1,1,1,1,1,1];
options = optimoptions('fsolve','Display','iter','MaxIterations',5000,'MaxFunEvals',5000);
[x,fval] = fsolve(equations,InitialGuess,options);
```

Mathematica Code Solution:

```
FindRoot[{158.49*(1.47+x4)-(1.47*((1.74+x2*x4)/0.23)^1.19) == 0,
(x1-26.3*x4)*(1.47+x4)-(x4*((1.74+x2*x4)/(x2*x3))^x2) == 0,
173.78*(1.76+x5)-(1.76*((2.09+x2*x5)/0.23)^1.19) == 0,
(x1-13.2*x5)*(1.76+x5)-(x5*((2.09+x2*x5)/(x2*x3))^x2) == 0,
186.21*(2.01+x6)-(2.01*((2.39+x2*x6)/0.23)^1.19) == 0,
(x1-5.4*x6)*(2.01+x6)-(x6*((2.39+x2*x6)/(x2*x3))^x2) == 0},
{{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)},
{x4, 10*(2*RandomReal[]-1)},
{x5, 10*(2*RandomReal[]-1)},
{x6, 10*(2*RandomReal[]-1)}}},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```

interface(displayprecision = 16)
with(RandomTools):
fsolve({158.49*(1.47+x4)-(1.47*((1.74+x2*x4)/0.23)^1.19) = 0,
(x1-26.3*x4)*(1.47+x4)-(x4*((1.74+x2*x4)/(x2*x3))^x2) = 0,
173.78*(1.76+x5)-(1.76*((2.09+x2*x5)/0.23)^1.19) = 0,
(x1-13.2*x5)*(1.76+x5)-(x5*((2.09+x2*x5)/(x2*x3))^x2) = 0,
186.21*(2.01+x6)-(2.01*((2.39+x2*x6)/0.23)^1.19) = 0,
(x1-5.4*x6)*(2.01+x6)-(x6*((2.39+x2*x6)/(x2*x3))^x2) = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1),
x4=10*(2*Generate(float)-1), x5=10*(2*Generate(float)-1), x6=10*(2*Generate(float)-1)})

```

The results for Case XI are shown in Table 15:

Table 15: Numerical solutions and residual errors for Case XI

|             | Parameter Values                                                                                                                      | Residual values                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1stOpt      | x1: -204.645852986762<br>x2: 1.18367346938673<br>x3: 0.190905481324504<br>x4: -1.47<br>x5: -1.75984340167139<br>x6: -2.00970507730151 | -7.22473161868278E-14<br>8.68286102959847E-14<br>4.98025232165134E-13<br>-5.40512079538757E-13<br>-7.50594031373453E-13<br>7.37195027245008E-13 |
| Matlab      | No correct solution                                                                                                                   |                                                                                                                                                 |
| Mathematica | No correct solution                                                                                                                   |                                                                                                                                                 |
| Maple       | No correct solutio                                                                                                                    |                                                                                                                                                 |

1stOpt is the only software that successfully finds the correct solution for this problem with nearly 100% success rate.

### 1.2.12 Case XII: A eight-equation nonlinear system with eight variables

$$x_1 \cdot \exp(-\exp(x_2 - x_3 \cdot (a_i - x_4)^2)) + \frac{x_5}{1 + x_6 \cdot \exp(x_7 \cdot a_i)} + x_8 = b_i, i = 1 \dots 8$$

where: a=[1,2,3,4,5,6,7,48]; b=[119, 163, 240, 395, 617, 1461, 4310, 36955]

Case XII - Code Solution

1stOpt Code Solution:

```
Hardness = 2;
Constant a=[1,2,3,4,5,6,7,48],b=[119,163,240,395,617,1461,4310,36955];
Function For(i=1:8,a,b)(x1*exp(-exp(x2-x3*(a-x4)^2))+x5/(1+x6*exp(x7*a))+x8=b);
```

Matlab Code Solution:

```
equations = @(x) [x(1)*exp(0-exp(x(2)-x(3)*(1-x(4))^2))+x(5)/(1+x(6)*exp(x(7)*1))+x(8)-119;
x(1)*exp(0-exp(x(2)-x(3)*(2-x(4))^2))+x(5)/(1+x(6)*exp(x(7)*2))+x(8)-163;
x(1)*exp(0-exp(x(2)-x(3)*(3-x(4))^2))+x(5)/(1+x(6)*exp(x(7)*3))+x(8)-240;
x(1)*exp(0-exp(x(2)-x(3)*(4-x(4))^2))+x(5)/(1+x(6)*exp(x(7)*4))+x(8)-395;
x(1)*exp(0-exp(x(2)-x(3)*(5-x(4))^2))+x(5)/(1+x(6)*exp(x(7)*5))+x(8)-617;
x(1)*exp(0-exp(x(2)-x(3)*(6-x(4))^2))+x(5)/(1+x(6)*exp(x(7)*6))+x(8)-1461;
x(1)*exp(0-exp(x(2)-x(3)*(7-x(4))^2))+x(5)/(1+x(6)*exp(x(7)*7))+x(8)-4310;
x(1)*exp(0-exp(x(2)-x(3)*(48-x(4))^2))+x(5)/(1+x(6)*exp(x(7)*48))+x(8)-36955];
InitialGuess = 10*(2*rand(8,1)-1); % [1,1,1,1,1,1,1,1];
options = optimoptions('fsolve','Display','iter','MaxIterations',5000,'MaxFunEvals',5000);
[x,fval] = fsolve(equations,InitialGuess,options);
```

Mathematica Code Solution:

```
FindRoot[{x1*Exp[0-Exp[x2-x3*(1-x4)^2]]+x5/(1+x6*Exp[x7*1])+x8-119 == 0,
x1*Exp[0-Exp[x2-x3*(2-x4)^2]]+x5/(1+x6*Exp[x7*2])+x8-163 == 0,
x1*Exp[0-Exp[x2-x3*(3-x4)^2]]+x5/(1+x6*Exp[x7*3])+x8-240 == 0,
x1*Exp[0-Exp[x2-x3*(4-x4)^2]]+x5/(1+x6*Exp[x7*4])+x8-395 == 0,
x1*Exp[0-Exp[x2-x3*(5-x4)^2]]+x5/(1+x6*Exp[x7*5])+x8-617 == 0,
x1*Exp[0-Exp[x2-x3*(6-x4)^2]]+x5/(1+x6*Exp[x7*6])+x8-1461 == 0,
x1*Exp[0-Exp[x2-x3*(7-x4)^2]]+x5/(1+x6*Exp[x7*7])+x8-4310 == 0,
x1*Exp[0-Exp[x2-x3*(48-x4)^2]]+x5/(1+x6*Exp[x7*48])+x8-36955 == 0},
{x1, 10*(2*RandomReal[]-1)},
{x2, 10*(2*RandomReal[]-1)},
{x3, 10*(2*RandomReal[]-1)},
{x4, 10*(2*RandomReal[]-1)},
{x5, 10*(2*RandomReal[]-1)},
{x6, 10*(2*RandomReal[]-1)},
{x7, 10*(2*RandomReal[]-1)},
{x8, 10*(2*RandomReal[]-1)}},
MaxIterations -> 100, WorkingPrecision -> 16]
```

Maple Code Solution:

```
interface(displayprecision = 16)
with(RandomTools):
fsolve({x1*exp(0-exp(x2-x3*(1-x4)^2))+x5/(1+x6*exp(x7*1))+x8-119 = 0,
x1*exp(0-exp(x2-x3*(2-x4)^2))+x5/(1+x6*exp(x7*2))+x8-163 = 0,
x1*exp(0-exp(x2-x3*(3-x4)^2))+x5/(1+x6*exp(x7*3))+x8-240 = 0,
x1*exp(0-exp(x2-x3*(4-x4)^2))+x5/(1+x6*exp(x7*4))+x8-395 = 0,
x1*exp(0-exp(x2-x3*(5-x4)^2))+x5/(1+x6*exp(x7*5))+x8-617 = 0,
x1*exp(0-exp(x2-x3*(6-x4)^2))+x5/(1+x6*exp(x7*6))+x8-1461 = 0,
x1*exp(0-exp(x2-x3*(7-x4)^2))+x5/(1+x6*exp(x7*7))+x8-4310 = 0,
x1*exp(0-exp(x2-x3*(48-x4)^2))+x5/(1+x6*exp(x7*48))+x8-36955 = 0},
{x1=10*(2*Generate(float)-1), x2=10*(2*Generate(float)-1), x3=10*(2*Generate(float)-1), x4
=10*(2*Generate(float)-1), x5=10*(2*Generate(float)-1), x6=10*(2*Generate(float)-1), x7
=10*(2*Generate(float)-1), x8=10*(2*Generate(float)-1)})
```

The results for Case XII are shown in Table 16:

Table 16: Numerical solutions and residual errors for Case XII

|             | Parameter Values       | Residual values       |
|-------------|------------------------|-----------------------|
| 1stOpt      | Solution 1:            | Solution 1:           |
|             | x1: -1143.06680965005  | 1.26476606965298E-12  |
|             | x2: 1.74650544185398   | -2.55795384873636E-13 |
|             | x3: 0.211000039051279  | 1.79056769411545E-12  |
|             | x4: 2.21121626657503   | 2.04636307898909E-12  |
|             | x5: 37981.2000329655   | 1.13686837721616E-12  |
|             | x6: 5189.00679471753   | -9.09494701772928E-13 |
|             | x7: -0.961716225777626 | -1.81898940354586E-12 |
|             | x8: 116.866776684532   | -2.18278728425503E-11 |
|             | Solution 2:            | Solution 2:           |
|             | x1: 876.600945722035   | -1.59161572810262E-12 |
|             | x2: -3.39977707570623  | 1.25055521493778E-12  |
|             | x3: -0.311164794133935 | -4.54747350886464E-13 |
|             | x4: 1.89055245389734   | 4.54747350886464E-13  |
|             | x5: 37700.2364638881   | 3.18323145620525E-12  |
|             | x6: 3865.61598289562   | 2.04636307898909E-12  |
|             | x7: -0.913515606842424 | 1.81898940354586E-12  |
|             | x8: -745.23646388806   | 1.45519152283669E-11  |
| Matlab      | No correct solution    |                       |
| Mathematica | No correct solution    |                       |
| Maple       | No correct solutio     |                       |

This problem is highly challenging. 1stOpt succeeds in returning one of the two correct solutions in 80% of the runs, whereas the three "3M" software consistently fails to obtain any correct result. Even when supplied with initial values close to the correct solution, such as  $x=[9000, -2.5, 0.5, 6, 37000, 3800, -1, -9000]$  or  $x=[-1200, 1.5, 0.3, 3.0, 35000, 6000, -0.5, 90]$ , they are still unable to converge to the correct solution.

### 1.3 Summary

Based on the comparative performance across the twelve test problems, 1stOpt exhibits a noticeably stronger overall robustness than the three ‘3M’ software packages. Nevertheless, this conclusion is naturally constrained by our current understanding and usage of the 3M tools. It is entirely possible that more effective commands, solver options, or modeling strategies exist that could further improve their performance. Readers who achieve better results using 3M or other software/methods are warmly encouraged to share their findings in the comment section or join the discussion in the Discord channel I created (<https://discord.gg/pyNyuQNBd>). I sincerely look forward to your insights and contributions.

Some readers may also question whether such a comparison is fully fair. Indeed, the nonlinear equation solvers in the 3M software are primarily based on local, quasi-Newton-type methods, whose convergence heavily depends on the quality of the initial guesses. Identifying suitable initial values is not only extremely challenging for ordinary users, but often difficult even for domain experts. Before the emergence of 1stOpt, it was widely accepted that users themselves had to adjust and refine initial guesses, as no algorithmic theory could guarantee convergence from arbitrary starting points—even global optimization algorithms offer no such guarantee.

However, the performance of 1stOpt suggests that the long-standing “era of guessing initial values” can, to a large extent, come to an end. For typical users, it is now sufficient to simply input the mathematical formulation of the problem; the software then autonomously searches for and returns the correct solution with high reliability. The underlying algorithmic complexity can be left to mathematicians and computer scientists. In this regard, 1stOpt represents a qualitative breakthrough. Although it cannot guarantee 100% accuracy from random initial values, its practical robustness is substantially higher than that of the 3M tools, especially for challenging nonlinear systems.

It is important to emphasize that Matlab, Mathematica, and Maple remain among the most powerful and versatile scientific computing platforms available today, with capabilities that extend far beyond nonlinear equation solving. The present comparison focuses solely on this specific task. Different solver options, parameter settings, or code constructions may lead to different outcomes, and therefore the conclusions of this article should be interpreted as a reference rather than a definitive ranking.

I have personally used 1stOpt for several years and have been consistently impressed by its performance in solving nonlinear equations, particularly in low-dimensional cases. Its ability to locate correct solutions with high accuracy and reliability is remarkable. By contrast, the 3M software—while outstanding in many other domains—often struggles with difficult nonlinear systems, especially when good initial guesses are required.

In the future, I plan to introduce additional test cases to further demonstrate the capabilities of 1stOpt not only in solving nonlinear equations but also in tackling other types of global optimization problems. I also intend to share more coding techniques and practical toolboxes in 1stOpt to help users better understand and utilize the software. I hope that more researchers and practitioners will explore 1stOpt for their nonlinear equation-solving needs, as it has the potential to significantly enhance both efficiency and accuracy.

If you have similar challenging nonlinear optimization problems or would like to discuss related topics, please feel free to join the Discord channel (<https://discord.gg/pyNyuQNBd>). I look forward to seeing you there.